

VK-RMD: Reliability-Guided Missing-Modality Distillation for Privacy-Friendly Multimodal Human Perception

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Abstract

Multimodal human perception is increasingly expected to operate in privacy-sensitive environments where raw RGB streams are undesirable at inference time and where sensing modalities may be intermittently unavailable. Existing multimodal human pose estimation (HPE) and human activity recognition (HAR) systems often rely on fixed input configurations; once depth, LiDAR, mmWave, WiFi-CSI, or visual-keypoint streams are missing or degraded, performance can collapse even when the full-modality model is accurate. This paper presents VK-RMD, a reliability-guided missing-modality distillation framework for privacy-friendly multimodal human perception. The central idea is to use visual keypoints (VKs) as a structural bridge: RGB information is used only as training-time privileged knowledge through VK-based supervision, while the deployable student performs inference from VK/non-RGB or fully non-visual modalities. The student is trained with random modality-subset sampling and a reliability-guided fusion module that dynamically weights available modalities under each missing pattern. HPE validates fine-grained body-structure estimation, and HAR validates action-level semantic recognition, giving an application-oriented evaluation of the same robustness mechanism. On MMFi-based HPE evaluation, VK-RMD Student-VK achieves 75.18 mm average MPJPE over 31 modality combinations, compared with 244.91 mm for the reproduced X-Fi/VK baseline; the non-visual Student-NV achieves 84.68 mm over 15 non-visual combinations. On HAR, Student-VK achieves 85.15% average accuracy over 15 combinations, compared with 69.63% for the reproduced baseline, while Student-NV achieves 80.62% over non-visual combinations. Cross-scene and cross-subject protocols further show that the proposed design remains effective

under practical distribution shifts. Ablations show that reliability-guided fusion improves robustness over uniform fusion in both HPE and HAR, while additional endpoint and reliability-weight analyses clarify the conditions under which each component is useful.

Keywords: Multimodal human perception, missing modalities, visual keypoints, human pose estimation, human activity recognition, reliability-guided fusion, knowledge distillation, privacy-friendly sensing

1. Introduction

Human pose estimation (HPE) and human activity recognition (HAR) are two complementary tasks in human-centered artificial intelligence. HPE estimates fine-grained body geometry, typically through 2D or 3D joint coordinates; HAR recognizes action semantics, such as sitting, walking, or interacting with an environment. Many practical systems need both capabilities: geometry is needed to understand posture and body structure, while activity labels summarize the semantic state of a human subject. The engineering challenge is not merely to achieve high accuracy under complete laboratory inputs. A robust system must continue to function when some sensing channels are absent, noisy, or unsuitable for privacy-sensitive inference.

Raw RGB images are highly informative, but they are also privacy-sensitive. They expose identity cues, clothing, body appearance, and background context. In many long-term monitoring or human-computer interaction settings, continuous RGB inference is difficult to justify. Non-RGB modalities such as depth, LiDAR, mmWave radar, and WiFi channel state information (WiFi-CSI) provide alternatives with reduced visual exposure. However, these modalities are heterogeneous and noisy: depth can be affected by range and occlusion, LiDAR point clouds may be sparse for human bodies, mmWave radar can be noisy and variable, and WiFi-CSI is indirect and environment-dependent. A model that simply concatenates all available sensors and assumes a fixed modality set is therefore fragile.

Visual keypoints (VKs) provide a useful intermediate representation. They retain explicit human-body structure while removing most raw appearance and background information. Prior X-Fi/VK-style work shows that replacing raw RGB with structured keypoint representations can reduce privacy risk and preserve pose-related information. Nevertheless, using VKs alone does not solve arbitrary missing-modality robustness. A VK stream may it-

self be unavailable or inaccurate, and practical multimodal perception must work with variable subsets of non-RGB sensors. This motivates a broader formulation: privacy-friendly multimodal human perception under missing modalities.

This paper proposes VK-RMD, a reliability-guided missing-modality distillation framework. The method treats RGB as training-time privileged knowledge rather than an inference input. VKs act as a structural bridge from visual knowledge to deployable multimodal sensing. The student model is trained by randomly sampling modality subsets and by learning to match stable supervision from full-modality teachers and ground-truth labels. During inference, the student receives whatever modalities are available and uses reliability-guided fusion to dynamically assign weights to modality-specific evidence.

The paper follows an engineering application perspective. We do not claim to introduce knowledge distillation, skeleton representations, or modality dropout as isolated inventions. Instead, we focus on an integrated, testable solution to a practical problem: how to build a privacy-friendly multimodal HPE/HAR model that does not require raw RGB at inference and remains usable under arbitrary modality combinations. The contribution is validated through extensive results rather than through a purely theoretical claim.

This distinction is important for the target venue. Engineering Applications of Artificial Intelligence papers are expected to connect an artificial-intelligence method to a real engineering need, to show why the design choices matter, and to demonstrate the method under realistic conditions. In this manuscript, the engineering need is robust human perception when privacy and sensor availability constraints prevent the use of a fixed RGB-centered pipeline. The artificial-intelligence component is not a single isolated module; it is a training and fusion strategy that combines VK-based structural supervision, missing-modality sampling, teacher-student learning, and reliability-guided decision weighting. The experimental evidence is therefore organized around application questions: Does the method work when all modalities are present? Does it remain usable when modalities are removed? Does it still work without VK? Does it generalize across scenes and subjects? Which component contributes most consistently?

The answer to these questions is mixed in a scientifically useful way. The proposed student clearly improves average all-combination robustness over reproduced X-Fi/VK baselines in both HPE and HAR. Reliability-guided

fusion is consistently better than uniform fusion, especially for HAR. Non-visual variants remain strong in full non-visual inference and acceptable in averaged missing-modality tests. At the same time, not every module should be oversold. HPE no-distillation is competitive, showing that distillation is not the sole source of improvement. The exploratory Super-Teacher is not strong enough under missing combinations to be a main contribution. These boundaries are explicitly stated because a credible engineering paper should report both the effective design decisions and the limitations that matter for deployment.

The main contributions are:

- We formulate VKs as a privacy-friendly structural bridge between RGB privileged training knowledge and deployable multimodal inference, distinguishing our setting from direct raw-RGB inference and from simple fixed-modality fusion.
- We propose a missing-modality student training strategy that samples modality subsets during training and evaluates all valid modality combinations during testing.
- We introduce reliability-guided fusion as an interpretable engineering mechanism for dynamically weighting available modalities under each missing pattern.
- We validate the same framework on HPE and HAR, covering random, cross-scene, cross-subject, VK-enabled, and non-visual settings.
- We provide ablation, complexity, robustness, and reliability-weight analyses, while clearly separating reproduced X-Fi/VK baselines from our own method.

2. Related work

2.1. Privacy-friendly human sensing and visual keypoints

RGB-based human perception has been widely studied because images provide dense visual cues. However, RGB inputs are often undesirable in privacy-sensitive human sensing. Visual keypoints and skeletons are commonly used as reduced representations of human structure. They preserve

joint-level geometry while suppressing identity, clothing, and background appearance. This reduction is not a formal privacy guarantee, but it lowers visual exposure compared with raw images.

The X-Fi/VK line of work is particularly relevant because it motivates the use of structured visual representations in multimodal human perception. In this paper, reproduced X-Fi/VK results are treated as baselines. We do not present X-Fi/VK itself as our contribution. Our contribution begins from the observation that a VK-based model still needs robustness to arbitrary missing modalities and dynamic sensor reliability.

Privacy-friendly human sensing can be approached from several directions. One direction removes RGB entirely and relies on RF, acoustic, inertial, depth, or point-cloud signals. This reduces visual exposure but may weaken pose precision or semantic recognition. A second direction keeps RGB during inference but applies blurring, anonymization, or representation learning; such methods can still be questioned because raw visual capture remains part of the pipeline. A third direction uses structured body representations such as skeletons or keypoints. The third direction is the most relevant to our work because it retains the geometry needed by HPE and HAR while avoiding direct raw-RGB inference. However, VK alone is not sufficient: keypoints may be missing, noisy, or unavailable, and a robust system still needs non-RGB alternatives.

The paper therefore uses a two-level privacy-friendly formulation. Student-VK uses VK together with non-RGB modalities and removes raw RGB. Student-NV removes VK as well and uses only non-visual modalities. Reporting both variants prevents an overconfident privacy claim. Student-VK demonstrates the benefit of VK as a structural bridge; Student-NV demonstrates the feasibility of a stricter non-visual inference setting.

2.2. Multimodal HPE and HAR

Multimodal HPE combines visual and non-visual signals to infer body joints. Multimodal HAR combines complementary sensing streams to classify actions. These tasks are often studied separately, yet they are naturally connected: pose estimates describe body structure, and action labels describe temporal or semantic behavior. A robust human perception model should be useful for both tasks. In this work, HPE and HAR are not two unrelated stories; they are two validation layers for the same privacy-friendly missing-modality framework.

The two tasks also stress different failure modes. HPE is sensitive to geometric errors and therefore exposes whether the model preserves body structure under missing modalities. HAR is sensitive to semantic separability and therefore exposes whether the fused representation captures action-level information. A method that performs well only on HAR may hide poor structural precision; a method that performs well only on HPE may not support practical activity understanding. By evaluating both tasks, the paper tests whether the same robustness mechanism can support fine-grained regression and categorical recognition.

This dual-task evaluation should not be confused with claiming a general-purpose human perception model. The current work evaluates two representative tasks available in the project pipeline. It does not claim to address tracking, forecasting, segmentation, identification, or interaction reasoning. This boundary keeps the contribution precise.

2.3. Missing-modality learning

Missing-modality learning addresses situations where a model cannot assume that all inputs are available. Common approaches include modality dropout, feature reconstruction, modality hallucination, mixture-of-experts fusion, and adaptive gating. Our setting differs from simple training-time dropout because the evaluation explicitly reports all valid modality combinations. For HPE, five modalities produce 31 non-empty combinations; for HAR, four modalities produce 15 non-empty combinations. This all-combination evaluation forces the model to handle both mild and severe missing patterns.

The all-combination protocol is demanding but useful. A method can appear robust if it is evaluated only under one or two manually selected missing cases. In contrast, exhaustive combination evaluation reveals whether robustness is broad or concentrated in easy cases. In our results, full-modality performance is strong, moderate missing patterns are generally robust, and single-modality cases are still difficult. This shape of results is more informative than a single average number because it identifies where the method is ready and where additional work is needed.

For the EAAI audience, the missing-modality problem is an engineering reliability problem. Sensors may be unavailable due to hardware failure, occlusion, privacy policy, environmental interference, or cost constraints. Training and testing only the full-modality configuration would overestimate

real-world utility. The proposed random subset training and all-combination evaluation directly address this reliability concern.

2.4. Reliability-aware multimodal fusion

Uniform fusion treats all modalities as equally informative. This assumption is rarely valid in heterogeneous sensing. Different modalities vary in quality across scenes, subjects, and missing patterns. Reliability-aware fusion estimates dynamic weights for available modalities and therefore provides a more flexible engineering mechanism. In VK-RMD, the reliability module outputs normalized weights over the available modality tokens. The exported reliability-weight CSVs are used in this manuscript to visualize how the learned fusion module reallocates decision weight under different missing-modality patterns.

Reliability modeling also improves interpretability at the engineering level. It does not make the neural network fully transparent, but it gives a mechanism for asking which modalities receive larger decision weight under different input subsets. This is valuable for diagnosing sensor failures and for explaining why a model may rely more on depth in one setting and more on mmWave or VK in another. In this draft, reliability evidence is supported by both performance ablations and exported average reliability weights.

2.5. Teacher-student distillation for robust perception

Teacher-student distillation transfers supervision from a stronger or more complete model to a student that is smaller, more deployable, or trained under restricted inputs. In our formulation, the teacher can use full-modality and VK/RGB-derived knowledge during training, while the final student is evaluated without raw RGB and under missing modalities. The HPE no-distillation ablation is strong; therefore, we interpret distillation cautiously. Distillation is one part of the privileged training chain, not a guaranteed source of improvement. The most consistent evidence in our results supports reliability-guided missing-modality learning, with distillation providing additional structure especially for HAR and for the overall training design.

This interpretation avoids a common writing trap. It would be tempting to describe the work as a distillation paper and claim that distillation alone explains the improvement. The actual evidence is more nuanced. HAR benefits from distillation in average robustness, while HPE shows that a no-distillation variant can be competitive. The correct story is therefore not “distillation always wins,” but “privileged supervision, missing-modality

Table 1: Capability-level comparison with related lines of work. Exact literature entries should be finalized after citation verification.

Method family	Privacy-friendly	HPE	HAR	Missing modality	Reliability modeling	Distillation	Remarks
RGB HPE/HAR	No	Yes	Yes	Usually no	Rare	Optional	Strong visual cues but privacy-sensitive
Skeleton/VK methods	Partly	Yes	Yes	Limited	Limited	Optional	Reduce appearance but often assume fixed inputs
Non-RGB sensing	Yes	Yes/No	Yes/No	Limited	Limited	Rare	Privacy-friendly but modality quality varies
Missing-modality learning	Task-dependent	Optional	Optional	Yes	Optional	Optional	Often not evaluated on both HPE and HAR
X-Fi/VK reproduced baseline	More private than RGB	Yes	Yes	Weak in our all-combination test	Limited	Baseline	Prior/reproduced baseline only
VK-RMD	Yes	Yes	Yes	Yes	Yes	Yes	All-combination HPE/HAR validation

training, and reliability-guided fusion together form a deployable robustness framework.” This phrasing better reflects the data and is more likely to withstand review.

Table 1 summarizes the position of VK-RMD relative to relevant method families. The table is intentionally organized by capability rather than by claimed superiority. This makes the distinction from prior VK/X-Fi and missing-modality learning clearer.

3. Problem formulation

3.1. Multimodal human perception setting

Let $\mathcal{M}_p = \{\text{VK}, D, L, R, W\}$ denote the HPE modality set, where D , L , R , and W denote depth, LiDAR, mmWave, and WiFi-CSI. Let $\mathcal{M}_a = \{\text{VK}, D, L, R\}$ denote the HAR modality set used by the current implementation. For each sample, only a subset $\mathcal{S} \subseteq \mathcal{M}$ may be available. The model receives $\mathbf{x}_\mathcal{S} = \{\mathbf{x}_m : m \in \mathcal{S}\}$ and must produce either a pose estimate or an action prediction.

For HPE, the target is a 3D pose $\mathbf{Y}^p \in \mathbb{R}^{J \times 3}$ with $J = 17$ joints. For HAR, the target is an action label $y^a \in \{1, \dots, C\}$, where $C = 27$ for the current MMFi action setting. The student model is written as

$$f_\theta(\mathbf{x}_\mathcal{S}, \mathcal{S}) = \begin{cases} \hat{\mathbf{Y}}^p, & \text{HPE,} \\ \hat{\mathbf{p}}^a, & \text{HAR.} \end{cases} \quad (1)$$

3.2. Privacy-friendly VK representation

RGB is not used by the deployable student. It is used only indirectly through VK supervision and teacher training. VK is represented by 17 joint coordinates. Compared with raw RGB, VK removes most appearance and background information. Compared with a purely non-visual setting, VK provides explicit structural cues. This makes VK a useful bridge between privileged visual knowledge and practical inference.

3.3. Missing-modality setting

Training samples modality subsets from a predefined distribution. Evaluation reports all valid combinations. This distinction is important: random subset training is a training strategy, while all-combination evaluation is a robustness test. The model succeeds only if it performs well across combinations, not merely in the full-modality case.

3.4. HPE and HAR objectives

HPE is evaluated using mean per-joint position error (MPJPE), Procrustes-aligned PA-MPJPE, and MSE. HAR is evaluated using accuracy and macro-F1 when available. These two tasks test complementary properties of the same framework: HPE measures structural precision, and HAR measures semantic recognition.

3.5. Engineering requirements

The framework is designed around five requirements. First, raw RGB should not be required at inference. Second, the model should support all non-empty modality subsets rather than a single fixed input format. Third, the fusion mechanism should adapt to the available modalities rather than assuming equal modality quality. Fourth, the same training principle should support both HPE and HAR. Fifth, the evaluation should include random, cross-scene, and cross-subject protocols so that performance is not limited to one split.

These requirements determine the design choices. VK provides a structural bridge for the first requirement. Random subset training and all-combination evaluation address the second. Reliability-guided fusion addresses the third. Task-specific HPE/HAR heads address the fourth. Cross-protocol evaluation addresses the fifth. This requirement-to-design mapping is also the logic behind Table 11.

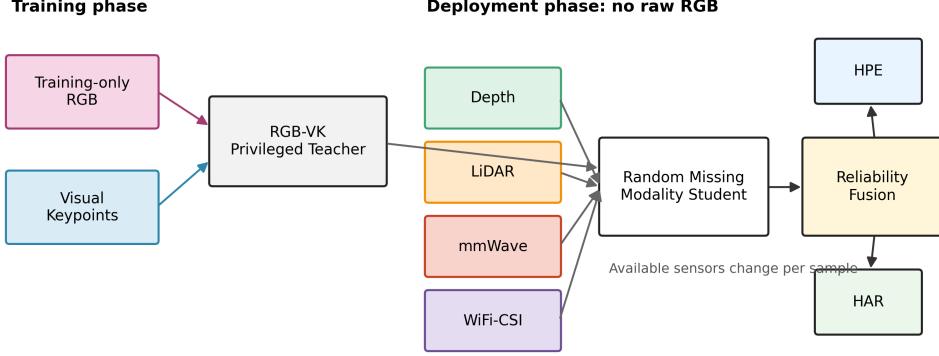


Figure 1: Overall architecture of VK-RMD. RGB is used only in the training phase through privileged VK-based supervision. During inference, no raw RGB stream is required; the student receives available VK/non-RGB or non-visual modalities, applies reliability-guided fusion, and predicts HPE and HAR outputs.

4. Proposed method

4.1. Overview of VK-RMD

Fig. 1 shows the overall architecture. The training phase uses RGB-derived VK and full-modality supervision to train a teacher/student framework. The deployment phase excludes raw RGB and processes available VK/non-RGB or purely non-visual modalities. The student is trained under random missing patterns and performs reliability-guided fusion before task-specific heads.

The figure should be read from left to right. The left side describes training knowledge: raw RGB is not a deployed modality, but its structural information is distilled through VK. The middle part describes the variable-modality student: different samples can contain different available sensors. The right side describes decision making: reliability fusion produces a fused representation for either HPE or HAR. This organization reflects the paper’s central claim that privacy-friendly inference and missing-modality robustness should be designed together.

4.2. Multimodal feature encoding

Each modality is encoded using a modality-specific extractor. VK is processed by a lightweight joint-aware encoder. The design avoids a large

flattened mapping: each 2D joint coordinate is lifted independently and then projected across joints to allow global interaction. Depth, LiDAR, mmWave, and WiFi-CSi features are encoded through corresponding modality backbones and projected into a shared token dimension.

Let $\phi_m(\cdot)$ denote the feature extractor for modality m . The encoded modality token is

$$\mathbf{z}_m = \phi_m(\mathbf{x}_m) + \mathbf{e}_m, \quad (2)$$

where $\mathbf{e}_m$ is a modality embedding. The embedding is necessary because the model receives variable modality subsets; it must know whether a token came from depth, LiDAR, mmWave, WiFi-CSi, or VK.

The modality-specific design also avoids forcing all sensor streams into the same raw preprocessing pipeline. Depth images, point clouds, radar points, WiFi-CSi sequences, and VK coordinates have different statistical properties. A shared encoder from raw inputs would either ignore these differences or require excessive preprocessing. Instead, VK-RMD uses specialized encoders and then aligns features at the token level. This design follows a practical engineering compromise: preserve modality-specific feature extraction while allowing shared fusion.

4.3. Random missing-modality training

Fig. 2 illustrates the training and evaluation protocol. At training time, modality subsets are randomly sampled. At evaluation time, all valid modality combinations are tested. This protocol turns missing-modality inference from a post-hoc stress test into a central training objective.

The expected training objective is

$$\min_{\theta} \mathbb{E}_{(\mathbf{x}, y)} \mathbb{E}_{\mathcal{S} \sim q(\mathcal{S})} [\mathcal{L}(f_{\theta}(\mathbf{x}_{\mathcal{S}}, \mathcal{S}), y)], \quad (3)$$

where $q(\mathcal{S})$ is the modality-subset sampling distribution.

The training distribution should cover both mild and severe missing cases. If training samples only full-modality and one-missing-modality cases, the model may still fail when two or three modalities are unavailable. Conversely, if training overemphasizes single-modality cases, full-modality performance may suffer. The current implementation uses random missing counts such as one, two, and three missing modalities for student training. The evaluation is stricter than the sampling process because it reports every valid modality combination, including combinations that may be rare during training.

Missing-modality protocol

	VK	Depth	LIDAR	mmWave	WiFi CSI
Full	on	on	on	on	on
Drop 1	on	on	off	on	on
Drop 2	on	off	on	off	on
Drop 3	off	on	off	on	off
Non-visual	off	on	on	on	on
Single sensor	off	off	off	on	off

Training randomly samples modality subsets; evaluation reports all valid combinations.

Figure 2: Missing-modality protocol. Training randomly samples modality subsets, while evaluation reports all valid combinations. The figure is conceptually correct but should be redrawn before final submission because some column labels are close to the top boundary.

4.4. Reliability-guided fusion

For a given available subset $\mathcal{S}$, the model estimates a reliability score for each modality token. The normalized reliability weight is

$$\alpha_m = \frac{\exp(g(\mathbf{z}_m))}{\sum_{k \in \mathcal{S}} \exp(g(\mathbf{z}_k))}, \quad m \in \mathcal{S}. \quad (4)$$

The fused token is

$$\mathbf{z}_{\text{fuse}} = \sum_{m \in \mathcal{S}} \alpha_m \mathbf{z}_m. \quad (5)$$

Uniform fusion is a special case with $\alpha_m = 1/|\mathcal{S}|$. The ablation in Section 6.4 directly tests whether learned reliability weights are useful.

From an engineering perspective, reliability fusion has two advantages. The first is performance: the model can downweight weak or missing-condition modalities. The second is diagnosability: exported alpha weights can reveal whether the model consistently depends on a modality or shifts its decision weight according to the input subset. The present manuscript includes both ablation evidence and exported alpha-weight heatmaps.

4.5. Teacher-student distillation

The teacher uses full-modality information and privileged VK/RGB-derived supervision during training. The student is trained under missing subsets.

For HPE, the student minimizes ground-truth pose regression and optional teacher pose matching:

$$\mathcal{L}_p = \|\hat{\mathbf{Y}}_s^p - \mathbf{Y}^p\|_2^2 + \lambda_p \|\hat{\mathbf{Y}}_s^p - \hat{\mathbf{Y}}_t^p\|_2^2 + \lambda_b \mathcal{L}_{\text{bone}}. \quad (6)$$

For HAR, the student uses cross-entropy and logit distillation:

$$\mathcal{L}_a = \text{CE}(\hat{\mathbf{p}}_s^a, y^a) + \lambda_a \text{KL}(\hat{\mathbf{p}}_s^a \parallel \hat{\mathbf{p}}_t^a). \quad (7)$$

In the final system, the teacher is not the deployed model. The teacher is a training mechanism that provides stable supervision when the student is trained under incomplete inputs. This distinction is important because the teacher itself may not perform well when directly evaluated under missing combinations. The student is the model intended for missing-modality inference.

4.6. Task-specific heads

The HPE head regresses a 17×3 pose from the fused representation. The HAR head predicts a 27-class action distribution. The shared robustness mechanism is therefore task-agnostic, while the final output heads are task-specific.

5. Experimental setup

5.1. Dataset and tasks

Experiments are based on MMFi-style multimodal human perception data following the X-Fi/VK setting. HPE uses VK, depth, LiDAR, mmWave, and WiFi-CSI. HAR uses VK, depth, LiDAR, and mmWave in the current implementation. The dataset contains multiple environments, subjects, and actions, enabling random split, cross-scene, and cross-subject evaluation.

5.2. Baselines and reproduced X-Fi/VK settings

The reproduced X-Fi/VK outputs are treated as baselines. They are not counted as our method. This distinction is important for fair writing: the baseline provides a reference for prior VK-based multimodal perception, while VK-RMD adds random missing-modality training and reliability-guided fusion for robust deployment.

5.3. Evaluation metrics

HPE reports MSE, MPJPE, and PA-MPJPE. The main tables emphasize MPJPE because it is directly interpretable in millimeters. HAR reports accuracy and macro-F1 when available. For missing-modality robustness, we report average performance over all combinations and group performance by the number of missing modalities.

5.4. Implementation details

Training follows the project setting with 1000 training iterations per epoch where specified, while evaluation uses full validation sets unless explicitly stated. Checkpoints are selected by validation performance. The non-visual HPE student uses depth, LiDAR, mmWave, and WiFi-CSI; the non-visual HAR student uses depth, LiDAR, and mmWave. The same reliability-guided fusion principle is used across VK-enabled and non-visual variants.

The reported values are taken from local result files under the project directory. HPE main results are taken from `student_vk_random_all_combinations.csv`, `student_nv_random_nonvisual_combinations.csv`, and corresponding cross-scene/cross-subject files. HAR main results are taken from the analogous HAR result files. Ablation results are taken from no-distillation and uniform-fusion result files. The reproduced X-Fi/VK baselines are taken from the `origin-XFI` outputs. This traceability is important because the manuscript contains many experimental numbers; any number not traceable to a result file is marked as missing rather than invented.

5.5. Figure organization

The figure order follows the paper logic. Fig. 1 introduces the training-to-inference chain. Fig. 2 defines the missing-modality protocol. Fig. 3 summarizes the main HPE/HAR performance. Fig. 4 shows robustness as missing severity increases. Fig. 5 supports the module analysis. Fig. 6 visualizes exported reliability weights for representative missing-modality conditions. This ordering avoids the common problem of placing all figures at the end without narrative function.

5.6. Result inclusion policy

Because the project contains many result files, the manuscript follows a strict inclusion policy. Formal HPE and HAR student results are used as the main evidence. Reproduced X-Fi/VK results are used only as baselines. Ablation results are used only to support specific component claims.

Super-Teacher results are used only as exploratory analysis. Extremely poor missing-combination results are not used to make the main method look worse or better; instead, they are summarized in limitation or appendix-level discussion when they reveal a meaningful failure mode.

This policy is necessary for two reasons. First, a paper with many experiments can become confusing if every output is presented with equal weight. The reader needs to know which results define the method, which results compare against prior work, and which results are exploratory. Second, EAAI reviewers are likely to judge whether the experimental evidence supports the stated engineering application. If Super-Teacher failure cases were placed in the main table, the main story would become ambiguous. If reproduced X-Fi/VK outputs were described as our own method, the contribution would be misleading. The current draft avoids both problems.

The formal main results are therefore: HPE Student-VK, HPE Student-NV, HAR Student-VK, and HAR Student-NV under random, cross-scene, and cross-subject protocols. The main ablations are: no-distillation and uniform fusion. Encoder ablations are acknowledged but not emphasized because some combinations collapse severely. Super-Teacher is discussed only as a training exploration.

5.7. Protocol interpretation

The random split evaluates whether the model can learn the overall mapping under the standard data partition. The cross-subject protocol evaluates whether the learned representation generalizes to unseen subjects. The cross-scene protocol evaluates whether it generalizes across environments. These protocols have different difficulty levels. The observed results show that cross-scene is the hardest for HPE and HAR averages, which is plausible because non-RGB signals can be strongly affected by room layout, sensor placement, and environmental reflection patterns. Cross-subject results are closer to random split, suggesting that the learned representation is less sensitive to subject identity than to environmental shift.

This interpretation should be included in the final paper because it turns raw numbers into engineering insight. The goal is not only to state that cross-scene performance is lower; the goal is to explain why this matters. A model intended for practical use should not be evaluated only on random split. Scene shift exposes whether sensor fusion has learned general body cues or environment-specific correlations. The results suggest that VK-RMD

Table 2: Main HPE results. Lower MPJPE is better. “Reproduced X-Fi/VK” is a baseline, not our contribution.

Method	Role	Inference modalities	RGB inference	Protocol	Combinations	Avg. MPJPE (mm)	Full MPJPE (mm)
Reproduced X-Fi/VK	Baseline	VK+D+L+R+W	No	Random	31	244.91	244.70
VK-RMD Student-VK	Ours	VK+D+L+R+W	No	Random	31	75.18	50.17
VK-RMD Student-NV	Ours	D+L+R+W	No	Random	15	84.68	51.03
VK-RMD Student-VK	Ours	VK+D+L+R+W	No	Cross-scene	31	142.02	71.71
VK-RMD Student-NV	Ours	D+L+R+W	No	Cross-scene	15	153.89	78.96
VK-RMD Student-VK	Ours	VK+D+L+R+W	No	Cross-subject	31	78.57	49.88
VK-RMD Student-NV	Ours	D+L+R+W	No	Cross-subject	15	90.74	52.45

improves robustness, but scene generalization remains a target for future work.

6. Results and analysis

6.1. Main HPE results

Table 2 reports HPE results. The reproduced X-Fi/VK baseline obtains 244.91 mm average MPJPE over 31 modality combinations. VK-RMD Student-VK obtains 75.18 mm, showing substantially stronger robustness under arbitrary modality combinations. Student-NV obtains 84.68 mm over 15 non-visual combinations, demonstrating that the method remains effective when VK is removed at inference.

The random-split Student-VK result shows the benefit of VK as a structural bridge. The Student-NV result is particularly important for privacy-friendly inference because it removes VK as well as raw RGB. Cross-scene results are weaker than random and cross-subject results, indicating that scene shift remains a major challenge.

Several observations follow from Table 2. First, the gap between the reproduced baseline and Student-VK is large in average all-combination MPJPE. This does not mean that the baseline is poor in full-modality inference; rather, it indicates that a model not trained for arbitrary missing combinations can be fragile when evaluated exhaustively. Second, Student-NV is only moderately worse than Student-VK in random split full-modality inference (51.03 mm versus 50.17 mm). This suggests that non-visual modalities contain enough complementary information when all are available. Third, cross-scene evaluation is much harder than cross-subject evaluation. Student-VK increases from 75.18 mm average MPJPE in random split to 142.02 mm in cross-scene evaluation, while cross-subject performance remains 78.57 mm. This pattern implies that environmental shift affects sensor distributions more strongly than subject identity shift in the current setting.

Table 3: Main HAR results. Higher accuracy is better.

Method	Role	Inference modalities	RGB inference	Protocol	Combinations	Avg. accuracy (%)	Full accuracy (%)
Reproduced X-Fi/VK	Baseline	VK+D+L+R	No	Random	15	69.63	77.70
VK-RMD Student-VK	Ours	VK+D+L+R	No	Random	15	85.15	96.56
VK-RMD Student-NV	Ours	D+L+R	No	Random	7	80.62	96.12
VK-RMD Student-VK	Ours	VK+D+L+R	No	Cross-scene	15	80.61	95.25
VK-RMD Student-NV	Ours	D+L+R	No	Cross-scene	7	76.70	95.41
VK-RMD Student-VK	Ours	VK+D+L+R	No	Cross-subject	15	85.30	96.12
VK-RMD Student-NV	Ours	D+L+R	No	Cross-subject	7	81.26	95.70

The HPE results also clarify the role of VK. VK helps average robustness, but the non-visual variant remains competitive enough to support a stricter privacy-friendly interpretation. Therefore, the paper should not be written as “VK is always required.” Instead, VK is a bridge and optional inference modality; non-visual inference is a separate supported setting.

Comparison with reproduced baseline.. The reproduced baseline is useful because it answers a practical question: what happens if a VK/X-Fi style model is evaluated under all missing combinations without the proposed missing-modality adaptation? The high average MPJPE indicates that full-modality success does not automatically transfer to missing-modality success. This is one of the paper’s central engineering messages. It also explains why the main comparison uses all-combination averages rather than only the full-modality row.

Non-visual setting.. The Student-NV setting is stricter than Student-VK because it removes VK at inference. In HPE, Student-NV obtains 84.68 mm average MPJPE and 51.03 mm full non-visual MPJPE. The full non-visual result is close to Student-VK full-modality performance, but the average is worse because severe missing cases are harder without VK. This distinction should be described carefully: non-visual inference is feasible, but VK still provides useful structural information when allowed.

6.2. Main HAR results

Table 3 reports HAR results. The reproduced X-Fi/VK baseline obtains 69.63% average accuracy over 15 combinations. VK-RMD Student-VK obtains 85.15% average accuracy and 96.56% full-modality accuracy. Student-NV obtains 80.62% average accuracy across non-visual combinations and 96.12% full non-visual accuracy.

HAR results support the application value of the framework beyond pose regression. They show that the same missing-modality robustness strategy also improves action recognition.

HAR results also reveal a different robustness profile from HPE. Full-modality accuracy is high for both Student-VK and Student-NV, indicating that action classes can be recognized well when sufficient non-visual evidence is available. The average over all combinations is lower because single-modality and severe-missing cases are difficult. The reproduced baseline average of 69.63% shows that fixed or insufficiently adapted fusion is not enough for all-combination HAR. Student-VK improves the average to 85.15%, and Student-NV still reaches 80.62%, which is important for inference settings without VK.

Cross-scene HAR remains strong in full modality: Student-VK obtains 95.25% full accuracy and Student-NV obtains 95.41%. However, the average across combinations drops to 80.61% and 76.70%, respectively. This suggests that full sensor availability can compensate for scene shift, but missing sensors make generalization harder. This is a useful engineering finding: robust deployment should not rely on a single sensor in unseen scenes.

Action recognition under non-visual sensing.. HAR Student-NV is important because action recognition can often be performed from motion and geometry cues rather than visual appearance. The 96.12% full non-visual accuracy indicates that depth, LiDAR, and mmWave together provide strong semantic information. The lower 80.62% average across non-visual combinations shows that the challenge is not full non-visual recognition but robust recognition when only a subset remains. This again supports the missing-modality focus of the paper.

Why HAR strengthens the paper.. If the paper reported only HPE, reviewers could interpret VK-RMD as a pose-specific engineering solution. HAR shows that the same robust fusion design also supports action semantics. Conversely, if the paper reported only HAR, the method would lack fine-grained body-structure validation. Reporting both tasks makes the work more convincing as a multimodal human perception framework.

6.3. Robustness under missing modalities

Table 4 groups results by the number of missing modalities. HPE Student-VK degrades from 50.17 mm with all modalities to 78.11 mm when three modalities are missing, and to 124.73 mm in single-modality inference. Student-NV follows a similar pattern. HAR Student-VK remains above 87% accuracy with two missing modalities, but drops to 70.00% in single-modality

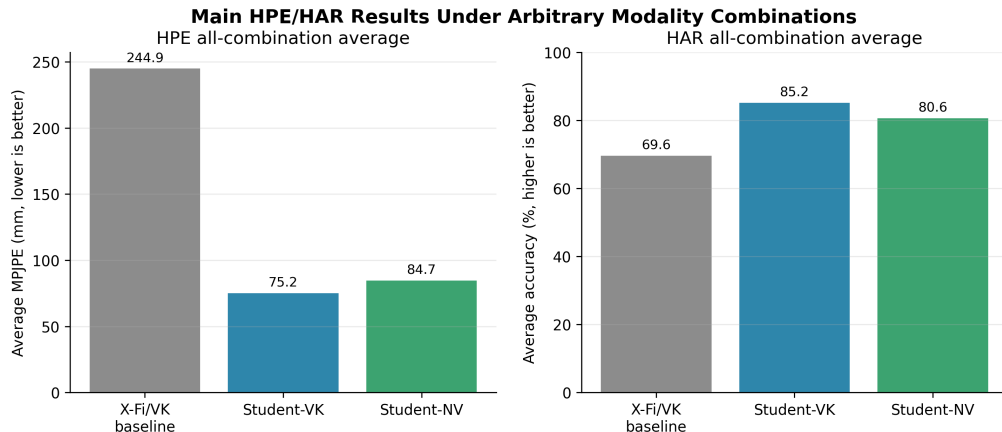


Figure 3: Main HPE and HAR results under arbitrary modality combinations. The left panel reports average HPE MPJPE over modality combinations; the right panel reports average HAR accuracy.

Table 4: Grouped missing-modality robustness. HPE reports MPJPE in mm; HAR reports accuracy in %.

Task/model	Metric	Missing 0	Missing 1	Missing 2	Missing 3	Missing 4
HPE Student-VK	MPJPE ↓	50.17	53.54	60.78	78.11	124.73
HPE Student-NV	MPJPE ↓	51.03	61.35	79.39	124.33	–
HAR Student-VK	Accuracy ↑	96.56	93.83	87.55	70.00	–
HAR Student-NV	Accuracy ↑	96.12	90.05	66.02	–	–

inference. HAR Student-NV drops more sharply when only one non-visual modality remains.

The degradation trend is expected: fewer modalities provide less complementary evidence. The key observation is not that single-modality inference is always strong, but that the trained student degrades gradually and remains substantially more robust than fixed full-modality baselines in all-combination evaluation.

The grouped analysis is more informative than reporting only the overall average. For HPE Student-VK, missing one modality increases MPJPE only from 50.17 mm to 53.54 mm. Missing two modalities increases it to 60.78 mm. This indicates that moderate missingness is well handled. The larger jump appears when only one or two modalities remain. For HAR Student-VK, missing one modality still gives 93.83% accuracy, and missing two modalities

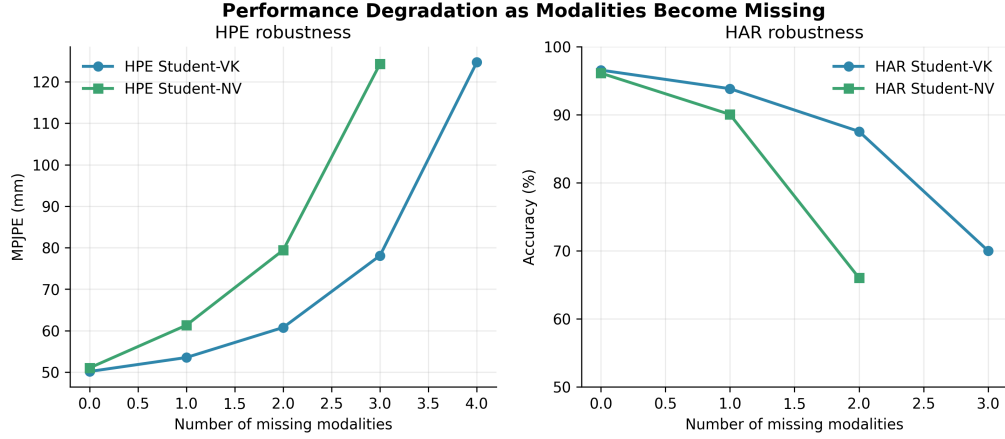


Figure 4: Performance degradation as modalities become missing. The curves summarize grouped missing-modality results from Table 4.

gives 87.55%. Again, severe missingness is the main remaining difficulty.

For Student-NV, the trend is similar but generally weaker in severe cases. This is expected because non-visual settings remove VK, one of the most structurally direct representations. The key value of Student-NV is therefore not that it always matches Student-VK, but that it provides a stricter non-visual inference option with strong full non-visual performance and reasonable all-combination averages.

Moderate missingness versus severe missingness.. The results suggest two regimes. In the moderate regime, one or two modalities are missing. Here, the model typically remains strong because complementary sensors are still available. In the severe regime, only one or two modalities remain. Performance decreases more sharply because each modality alone captures only part of the human state. This distinction is important for application planning. The proposed method can provide strong robustness for common missing cases, while severe single-modality scenarios may require additional modality-specific enhancement or confidence-based rejection.

Failure modes.. The most difficult combinations are those relying on weak single modalities. WiFi-CSI and LiDAR alone can be challenging for precise HPE. LiDAR-only HAR is also weak in several protocols. These failures are not hidden in the project results, but they are not placed in the main abstract-level claims because they represent expected edge cases. A final

Table 5: Ablation results. HPE reports average MPJPE over 31 combinations; HAR reports average accuracy over 15 combinations.

Task	Setting	Avg. metric	Full metric	Interpretation
HPE	VK-RMD Student-VK	75.18 mm	50.17 mm	Main model
HPE	w/o distillation	72.68 mm	48.83 mm	HPE KD gain is not consistent across all metrics; claim cautiously
HPE	Uniform fusion	82.77 mm	52.28 mm	Reliability fusion improves robustness
HPE	Simple MLP VK encoder	503.74 mm	77.69 mm	Severe missing-combination degradation; appendix-level detail
HPE	Skeleton prompt encoder	1000.80 mm	55.07 mm	Severe missing-combination degradation; appendix-level detail
HAR	VK-RMD Student-VK	85.15%	96.56%	Main model
HAR	w/o distillation	82.86%	95.99%	Distillation modestly improves average robustness
HAR	Uniform fusion	67.46%	95.39%	Reliability fusion is critical under missing modalities

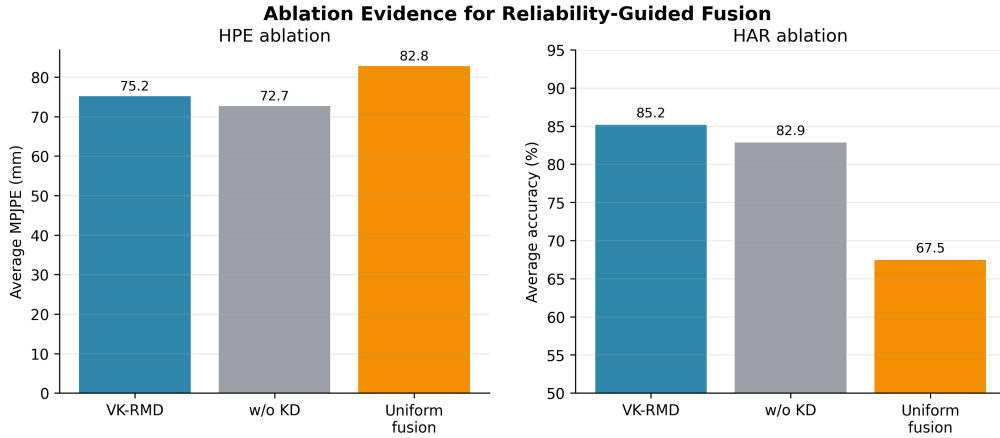


Figure 5: Ablation evidence for reliability-guided fusion and distillation. HPE no-distillation is competitive, so the manuscript does not overstate distillation; the reliability-fusion evidence is stronger and more consistent, especially for HAR.

submission may include a supplementary table of worst-case combinations to show transparency.

6.4. Ablation study

Table 5 summarizes ablation results. For HPE, uniform fusion increases average MPJPE from 75.18 mm to 82.77 mm. For HAR, uniform fusion reduces average accuracy from 85.15% to 67.46%. This strong HAR gap shows that reliability-guided fusion is not merely a cosmetic module; it is essential when modalities have different quality and availability.

The ablation results lead to two important writing decisions. First, reliability-guided fusion should be presented as a central and empirically supported component. The performance drop under uniform fusion is clear, especially in HAR. Second, distillation should be presented as part of the

Table 6: HPE distillation-variant endpoint results under full-modality evaluation. Lower is better.

Variant	MPJPE (mm)	PA-MPJPE (mm)	MSE
No KD	48.83	32.51	0.001396
Output KD	47.62	32.15	0.001365
Output+token KD	48.76	33.24	0.001408
Output+bone KD	49.34	33.38	0.001417
Full structural KD	49.20	34.63	0.001422

training framework rather than the only explanation for performance. The HPE no-distillation result is slightly better than the full model in the current all-combination average, which prevents a broad claim that distillation always improves HPE. This does not invalidate the framework, but it requires careful writing.

For EAAI, this kind of nuanced ablation is acceptable and even useful. Engineering application papers do not need every module to improve every metric monotonically; they need a clear explanation of which design choices are essential, which are task-dependent, and which require further study. In our case, reliability fusion is essential, missing-modality training is central, and distillation is useful but should be interpreted task-dependently.

Table 6 further breaks down the HPE distillation variants using full-modality endpoint evaluation. Output-level KD gives the lowest full-modality MPJPE among these variants (47.62 mm), while token and bone variants do not monotonically improve the full-modality endpoint. This result is reported to make the distillation claim precise: KD is part of the training design and can improve specific settings, but the strongest and most consistent robustness evidence comes from missing-modality training and reliability-guided fusion.

Table 7 summarizes additional full-modality endpoint results from the final checkpoints. These values are not a replacement for all-combination robustness metrics, but they help verify that the trained models did not trade away normal full-input performance. For HPE, Student-VK and Student-NV remain close to the full-modality teacher in the random split. For HAR, Student-VK and Student-NV slightly exceed the full-modality teacher in random-split endpoint accuracy, while cross-scene and cross-subject endpoint results remain above 95% for the student variants.

Table 7: Full-modality endpoint sanity checks from final checkpoints. HPE uses MPJPE; HAR uses accuracy. These endpoint values complement, but do not replace, all-combination evaluation.

Task	Model/protocol	Main metric	Secondary	Loss/MSE	Task	Model/protocol	Main metric	Secondary	Loss/MSE
HPE	Teacher, random	48.45 mm	33.06 mm	0.001392	HAR	Teacher, random	95.57%	95.56% F1	0.2033
HPE	Baseline full, random	49.14 mm	33.48 mm	0.001415	HAR	Student-VK, random	96.54%	96.53% F1	0.1627
HPE	Student-VK, random	50.17 mm	33.48 mm	0.001459	HAR	Student-NV, random	96.12%	96.13% F1	0.1939
HPE	Student-NV, random	51.03 mm	35.69 mm	0.001493	HAR	w/o distillation	95.97%	95.95% F1	0.2089
HPE	Teacher, cross-scene	76.06 mm	37.89 mm	0.002501	HAR	Uniform fusion	95.40%	95.40% F1	0.1873
HPE	Student-VK, cross-scene	71.70 mm	47.74 mm	0.002393	HAR	Student-VK, cross-scene	96.12%	96.10% F1	0.3624
HPE	Student-VK, cross-subject	49.88 mm	32.35 mm	0.001230	HAR	Student-NV, cross-scene	95.41%	95.40% F1	0.3327
HPE	Student-NV, cross-subject	52.46 mm	34.13 mm	0.001367	HAR	Student-NV, cross-subject	95.66%	95.64% F1	0.3926

Table 8: Representative exported reliability weights for Student-VK. Values are averaged over validation samples. Missing entries indicate unavailable modalities.

Task	Modality set	VK	Depth	LiDAR	mmWave	WiFi-CSI
HPE	VK+Depth	0.396	0.604	–	–	–
HPE	Depth+LiDAR	–	0.622	0.378	–	–
HPE	LiDAR+mmWave	–	–	0.489	0.511	–
HPE	VK+Depth+LiDAR	0.288	0.436	0.276	–	–
HPE	VK+Depth+LiDAR+mmWave+WiFi-CSI	0.189	0.270	0.178	0.183	0.179
HAR	VK+mmWave	0.414	–	–	0.586	–
HAR	LiDAR+mmWave	–	–	0.410	0.590	–
HAR	VK+Depth+mmWave	0.317	0.331	–	0.352	–
HAR	VK+Depth+LiDAR+mmWave	0.238	0.249	0.245	0.268	–

6.5. Reliability-guided fusion analysis

Reliability-guided fusion is supported by both the uniform-fusion ablation and the exported reliability weights. Fig. 6 visualizes the average normalized weights assigned to each available modality for HPE Student-VK and HAR Student-VK. The heatmap should not be interpreted as a complete explanation of the neural model, but it verifies that the fusion module does not collapse into fixed uniform averaging.

Table 8 reports representative averages from the same CSV files. For HPE, the model assigns more weight to depth than VK in the VK+Depth pair (0.604 versus 0.396), and also favors depth in Depth+LiDAR (0.622 versus 0.378). In the full five-modality HPE setting, weights are more distributed, with depth remaining the largest contributor (0.270). For HAR, mmWave receives larger weights than VK or LiDAR in combinations containing mmWave, such as VK+mmWave (0.586) and LiDAR+mmWave (0.590). These patterns are consistent with the performance tables, where depth and mmWave often provide strong complementary evidence. The analysis supports the engineering claim that reliability-guided fusion adapts to the available sensor subset.

Learned Reliability Weights Under Missing-Modality Conditions

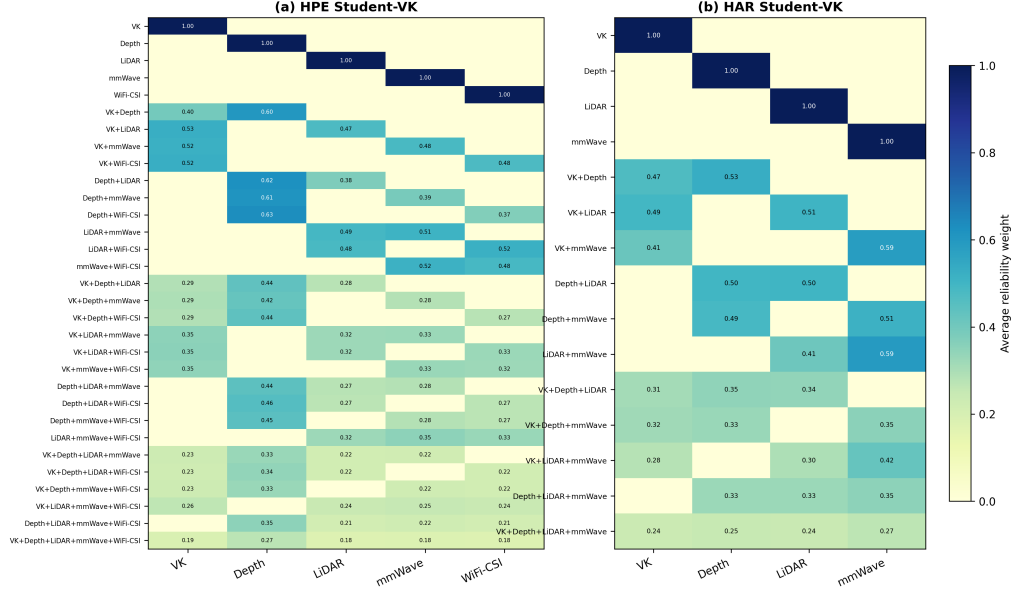


Figure 6: Learned reliability weights under missing-modality conditions. Rows indicate available modality combinations and columns indicate modalities. Blank cells indicate unavailable modalities. The exported weights show that VK-RMD dynamically reallocates decision weight according to the current modality subset rather than using a fixed uniform rule.

6.6. Noise and keypoint-drop robustness

The HPE Student-VK model was also evaluated under VK coordinate noise and joint dropping. With no perturbation, full-modality MPJPE is 50.18 mm. With 50% joint dropping and no coordinate noise, MPJPE increases to 54.75 mm. With coordinate noise standard deviation 20.0 and no dropping, MPJPE is 53.63 mm. With both noise standard deviation 20.0 and 50% joint dropping, MPJPE is 54.78 mm. These results indicate that the model does not rely exclusively on noise-free VK coordinates when other modalities are available.

The noise test also clarifies the role of VK. VK is useful, but the model is not simply copying VK coordinates. When VK is partially dropped or perturbed, other modalities can compensate. This is exactly the behavior expected from a multimodal student. However, the test is performed under full modality availability; it does not prove robustness under simultaneous

Table 9: HPE Student-VK robustness to VK noise and joint dropping.

Noise std.	Drop ratio	MPJPE (mm)	PA-MPJPE (mm)
0.0	0.0	50.18	33.48
0.0	0.5	54.75	35.50
10.0	0.0	52.08	34.57
20.0	0.0	53.63	35.20
20.0	0.5	54.78	35.54

Table 10: Measured HPE model complexity.

Model	Params	FPS	Peak memory (GB)
Student-VK	8.90M	62.01	4.95
Student-NV	8.90M	63.37	4.95
Baseline full	5.74M	52.05	4.94

VK corruption and sensor missingness. A stronger future experiment would combine VK noise with missing non-RGB modalities.

6.7. Complexity and deployability discussion

Table 10 reports measured complexity for HPE models. Student-VK and Student-NV each contain 8.90M parameters and run at approximately 62.01 FPS and 63.37 FPS in the measured environment. These numbers should be interpreted as implementation-specific measurements rather than hardware-independent performance.

The complexity table has two purposes. First, it prevents the paper from being evaluated only as an accuracy report. Second, it supports the claim that the student is not an excessively large research-only model. The reproduced baseline has fewer parameters but lower measured FPS, showing that parameter count alone does not determine throughput. A complete final submission should add HAR complexity and, if possible, FLOPs or latency measured under the same hardware and batch settings.

6.8. Summary of experimental evidence

Table 11 maps experiments to contributions. This table is useful for an EAAI submission because it makes clear how each engineering claim is supported by data.

Table 11: Experimental evidence supporting each claim.

Claim	Evidence	Boundary
Privacy-friendly inference without raw RGB	VK-enabled and non-visual HPE/HAR students	VK is privacy-friendly, not formally anonymous
Arbitrary missing-modality robustness	HPE 31 combinations; HAR 15 combinations; grouped missing results	Single-modality cases remain difficult
Reliability-guided fusion is useful	HPE/HAR uniform-fusion ablation; exported reliability-weight heatmaps	Weight analysis is diagnostic rather than a full causal explanation
HPE and HAR are jointly supported	Main HPE and HAR results under random, cross-scene, cross-subject protocols	They are two validation tasks, not a general task suite
Prior X-Fi/VK distinction is clear	Reproduced X-Fi/VK baseline is labeled baseline only	Baseline implementation details must be fully documented

This evidence table is also a guardrail for revision. If a claim is not supported by a row in Table 11, it should not be promoted as a main contribution. For example, the paper can claim reliability-guided fusion is useful because ablation supports it. The paper should not claim that Super-Teacher is a superior missing-modality model because the current exploration does not support it. This discipline is especially important when the project contains many experiments and tempting side stories.

7. Discussion

7.1. Relationship to X-Fi/VK prior work

The manuscript should be clear that X-Fi/VK is prior work and reproduced baseline. Our method builds on the insight that VK can reduce raw visual exposure while preserving human-body structure. The new contribution is the missing-modality training and reliability-guided fusion framework

that turns VK/non-RGB sensing into an all-combination robust HPE/HAR system.

The difference can be stated in three layers. At the representation layer, prior VK work motivates replacing raw RGB with structural keypoints. At the training layer, VK-RMD uses random modality subsets so that the student learns missing patterns rather than encountering them only at test time. At the fusion layer, VK-RMD uses reliability-guided weighting rather than fixed fusion. These layers jointly define the contribution and keep the paper from sounding like a simple reimplementation of X-Fi/VK.

7.2. Practical value for privacy-friendly human sensing

The practical value of VK-RMD lies in three properties. First, raw RGB is not required by the final student. Second, non-visual variants can operate without VK, using only depth, LiDAR, mmWave, and WiFi-CSI for HPE or depth, LiDAR, and mmWave for HAR. Third, the model is explicitly evaluated under missing modalities rather than assuming full sensors. These properties align with engineering application requirements: robustness, privacy-friendliness, and measurable task performance.

A second practical value is result transparency. Many multimodal papers report only the best full-modality configuration. In contrast, this work reports averages across all combinations, grouped missing-severity results, and non-visual variants. This exposes weak settings rather than hiding them. For an engineering application journal, such transparency is useful because it helps practitioners decide which modality sets are reliable and which require caution.

7.3. Limitations

The work has several limitations. First, the privacy claim is limited to reduced visual exposure; it is not a formal privacy guarantee. Second, single-modality performance remains weak for some modalities, especially LiDAR-only and WiFi-only settings. Third, cross-scene generalization is harder than random and cross-subject evaluation. Fourth, HPE distillation results are mixed, so distillation should not be overstated as the main source of improvement. Fifth, the reliability-weight heatmaps are diagnostic averages over exported fusion weights; they help explain fusion behavior but should not be interpreted as a causal proof of sensor quality.

These limitations do not invalidate the contribution, but they define the correct scope. The method is suitable for robustness-oriented multimodal

human perception, not for identity protection with formal guarantees. It is strong for full and moderate missing conditions, but not a complete solution for every single-sensor case. It improves average robustness, but it does not remove the need for scene adaptation. Stating these boundaries will make the paper more credible to reviewers.

7.4. Future work

Future work should improve single-modality robustness, study domain adaptation for cross-scene shifts, and connect reliability weights with calibrated uncertainty estimates. A more efficient and better-regularized Super-Teacher could be explored, but it should remain secondary unless it demonstrably improves downstream students under missing modalities.

7.5. Suitability for Engineering Applications of Artificial Intelligence

This manuscript is positioned for EAAI because it emphasizes an AI method applied to a concrete engineering problem: robust privacy-friendly human perception under incomplete multimodal sensing. The paper does not depend on a fabricated application scenario. Instead, it evaluates practical constraints directly through non-RGB inference, missing-modality combinations, cross-scene protocols, cross-subject protocols, complexity measurements, and ablations. The strongest EAAI-facing message is that the method transforms a fixed-modality human perception pipeline into a tested all-combination robust framework.

To further improve fit, the final revision should ensure that the introduction states the engineering problem before the algorithm, that every figure supports the application logic, and that the discussion explains deployment boundaries. The paper should also avoid overclaiming theory. Its strength is not a new mathematical learning principle; its strength is a carefully evaluated multimodal AI framework with application-oriented robustness evidence.

8. Conclusion

This paper presented VK-RMD, a reliability-guided missing-modality distillation framework for privacy-friendly multimodal human perception. The method uses VKs as a structured bridge from RGB privileged knowledge to non-RGB inference and trains a student model with random modality

subsets. HPE and HAR experiments demonstrate that the same framework supports body-structure estimation and action recognition under arbitrary modality combinations. Compared with reproduced X-Fi/VK baselines, VK-RMD substantially improves average all-combination robustness while maintaining strong full-modality performance. Ablation results show that reliability-guided fusion is an important component, especially for HAR. The paper deliberately avoids overstating distillation or Super-Teacher exploration and instead focuses on the validated engineering contribution: robust privacy-friendly HPE/HAR under missing modalities.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this draft, AI-assisted writing tools were used to reorganize manuscript structure, convert project notes into English academic prose, and format LaTeX tables. All experimental values should be verified by the authors against the original result files before submission.

Data availability

The experiments are based on MMFi-style data and local result files under the project directory. Public dataset links, code release links, and exact splits should be inserted before submission. [TO BE FILLED: Add official MMFi dataset URL, X-Fi repository URL, and code availability statement.]

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. [TO BE FILLED: Confirm before submission.]

Appendix A. Full HPE modality-combination results

Table A.12 lists the complete 31-combination HPE results for Student-VK. Table A.13 lists the complete 15-combination HPE results for Student-NV. Values are reported in millimeters.

Table A.12: Full HPE Student-VK results over all 31 modality combinations.

Modality set	MPJPE	PA-MPJPE	Modality set	MPJPE	PA-MPJPE	Modality set	MPJPE	PA-MPJPE
VK	92.66	42.60	Depth+LiDAR	53.00	35.35	VK+Depth+LiDAR	51.10	34.16
Depth	53.88	35.61	Depth+mmWave	52.32	34.64	VK+Depth+mmWave	50.50	33.52
LiDAR	140.14	93.52	Depth+WiFi	53.54	35.52	VK+Depth+WiFi	51.63	34.29
mmWave	114.98	58.13	LiDAR+mmWave	97.52	55.08	VK+LiDAR+mmWave	66.08	38.77
WiFi	222.14	110.19	LiDAR+WiFi	134.16	93.18	VK+LiDAR+WiFi	70.73	41.76
VK+Depth	51.65	34.29	mmWave+WiFi	108.54	57.27	VK+mmWave+WiFi	66.69	38.80
VK+LiDAR	73.18	41.92	Depth+LiDAR+mmWave	51.92	34.50	Depth+LiDAR+WiFi	52.85	35.38
VK+mmWave	68.93	38.90	Depth+mmWave+WiFi	52.13	34.62	LiDAR+mmWave+WiFi	94.17	55.05
VK+WiFi	88.11	42.30	VK+Depth+LiDAR+mmWave	50.25	33.45	VK+Depth+LiDAR+WiFi	51.05	34.18
VK+Depth+mmWave+WiFi	50.41	33.54	VK+LiDAR+mmWave+WiFi	64.25	38.75	Depth+LiDAR+mmWave+WiFi	51.78	34.54
VK+Depth+LiDAR+mmWave+WiFi	50.17	33.48	–	–	–	–	–	–

Table A.13: Full HPE Student-NV results over all 15 non-visual modality combinations.

Modality set	MPJPE	PA-MPJPE	Modality set	MPJPE	PA-MPJPE	Modality set	MPJPE	PA-MPJPE
Depth	52.34	36.44	Depth+LiDAR	52.03	36.39	Depth+LiDAR+mmWave	51.11	35.75
LiDAR	123.10	84.16	Depth+mmWave	51.24	35.86	Depth+LiDAR+WiFi	52.02	36.37
mmWave	109.44	57.49	Depth+WiFi	52.46	36.36	Depth+mmWave+WiFi	51.22	35.77
WiFi	212.40	106.87	LiDAR+mmWave	91.13	54.08	LiDAR+mmWave+WiFi	90.93	54.14
LiDAR+WiFi	123.65	84.02	mmWave+WiFi	106.06	57.44	Depth+LiDAR+mmWave+WiFi	51.03	35.69

Appendix B. Full HAR modality-combination results

Tables B.14 and B.15 list the complete random-split HAR results for Student-VK and Student-NV.

Table B.14: Full HAR Student-VK results over all 15 modality combinations.

Modality set	Acc.	Macro-F1	Modality set	Acc.	Macro-F1	Modality set	Acc.	Macro-F1
VK	65.31	65.70	VK+Depth	89.51	89.36	VK+Depth+LiDAR	89.62	89.49
Depth	88.02	87.86	VK+LiDAR	68.54	69.05	VK+Depth+mmWave	96.56	96.55
LiDAR	40.80	40.10	VK+mmWave	92.75	92.82	VK+LiDAR+mmWave	92.73	92.81
mmWave	85.88	86.06	Depth+LiDAR	88.52	88.39	Depth+LiDAR+mmWave	96.44	96.43
Depth+mmWave	96.45	96.44	LiDAR+mmWave	89.51	89.57	VK+Depth+LiDAR+mmWave	96.56	96.55

Appendix C. Reproduced X-Fi/VK baseline details

Tables C.16 and C.17 report the reproduced X-Fi/VK baseline outputs used for comparison. These results are not counted as contributions of VK-RMD; they provide the prior-work reference point for evaluating missing-modality robustness.

Appendix D. Exploratory Super-Teacher outputs

Table D.18 summarizes the Super-Teacher exploration. These results are intentionally kept outside the main contribution because the Super-Teacher itself is weak under missing-modality combination testing. The useful conclusion is methodological: full-modality or cross-task teaching does not automatically produce a missing-modality-robust inference model; the student-side missing-modality adaptation remains necessary.

Table B.15: Full HAR Student-NV results over all 7 non-visual modality combinations.

Modality set	Acc.	Macro-F1	Modality set	Acc.	Macro-F1
Depth	88.57	88.47	Depth+LiDAR	88.68	88.56
LiDAR	26.12	25.01	Depth+mmWave	96.18	96.19
mmWave	83.37	83.57	LiDAR+mmWave	85.30	85.38
Depth+LiDAR+mmWave	96.12	96.13	–	–	–

Table C.16: Reproduced X-Fi/VK baseline HPE results over 31 modality combinations.

Modality set	MPJPE	PA-MPJPE	Modality set	MPJPE	PA-MPJPE	Modality set	MPJPE	PA-MPJPE
VK	246.82	103.94	Depth	246.49	103.40	LiDAR	244.94	103.40
mmWave	244.24	103.41	WiFi	244.63	104.03	VK+Depth	246.62	103.57
VK+LiDAR	245.41	103.41	VK+mmWave	245.09	103.50	VK+WiFi	245.32	103.57
Depth+LiDAR	245.02	103.46	Depth+mmWave	244.12	103.55	Depth+WiFi	244.57	103.41
LiDAR+mmWave	244.82	103.41	LiDAR+WiFi	244.34	103.55	mmWave+WiFi	244.32	103.61
VK+Depth+LiDAR	245.32	103.39	VK+Depth+mmWave	245.15	103.31	VK+Depth+WiFi	245.32	103.45
VK+LiDAR+mmWave	245.15	103.52	VK+LiDAR+WiFi	244.50	103.69	VK+mmWave+WiFi	244.20	103.55
Depth+LiDAR+mmWave	244.37	103.46	Depth+LiDAR+WiFi	244.35	103.45	Depth+mmWave+WiFi	244.72	103.44
LiDAR+mmWave+WiFi	244.35	103.69	VK+Depth+LiDAR+mmWave	245.03	103.32	VK+Depth+LiDAR+WiFi	244.64	103.52
VK+Depth+mmWave+WiFi	244.95	103.43	VK+LiDAR+mmWave+WiFi	244.12	103.63	Depth+LiDAR+mmWave+WiFi	244.39	103.46
VK+Depth+LiDAR+mmWave+WiFi	244.70	103.53	–	–	–	–	–	–

Appendix E. Result-file coverage

Table E.19 records how local outputs are used in the manuscript. This table is included to make the large experimental base auditable and to prevent auxiliary or poor exploratory results from being confused with the formal claims.

Appendix F. Additional notes on figure readiness

The six inserted figures are current manuscript figures rather than final camera-ready artwork. Figs. 1, 3, 4, 5, and 6 are usable for the present draft. Fig. 2 should be redrawn because some labels are close to the boundary. The reliability heatmap is based on exported CSV weights and should be retained unless a cleaner camera-ready redraw is prepared.

Appendix G. References to verify

References

- [1] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” arXiv:1503.02531, 2015.
- [2] [\[SOURCE TO BE VERIFIED: Learning using privileged information reference.\]](#)

Table C.17: Reproduced X-Fi/VK baseline HAR results over 15 modality combinations. Macro-F1 was not exported by this baseline script, so only accuracy and cross-entropy loss are reported.

Modality set	Acc.	Loss	Modality set	Acc.	Loss	Modality set	Acc.	Loss
VK	75.93	0.697	Depth	45.61	4.391	LiDAR	39.83	1.969
mmWave	80.78	0.672	VK+Depth	56.32	2.386	VK+LiDAR	77.29	0.663
VK+mmWave	91.66	0.254	Depth+LiDAR	47.71	3.738	Depth+mmWave	69.04	1.327
LiDAR+mmWave	84.54	0.517	VK+Depth+LiDAR	58.58	2.115	VK+Depth+mmWave	77.09	0.828
VK+LiDAR+mmWave	91.47	0.257	Depth+LiDAR+mmWave	70.87	1.232	VK+Depth+LiDAR+mmWave	77.70	0.796

Table D.18: Exploratory Super-Teacher results. These are appendix-level diagnostic results, not main performance claims.

Model/evaluation	Average metric	Full metric
Super-Teacher HPE, 31 combinations	1538.81 mm	59.78 mm
Super-Teacher HAR, 15 combinations	53.59%	90.55%
Super-HPE-Student-VK, 31 combinations	82.06 mm	56.82 mm
Super-HPE-Student-NV, endpoint	—	59.71 mm

[Cao et al.] [\[SOURCE TO BE VERIFIED: OpenPose / part affinity fields reference.\]](#)

[Yan et al.] [\[SOURCE TO BE VERIFIED: Spatial temporal graph convolutional networks for skeleton-based action recognition.\]](#)

[MMFi] [\[SOURCE TO BE VERIFIED: Official MMFi dataset paper and GitHub repository.\]](#)

[X-Fi] [\[SOURCE TO BE VERIFIED: Original X-Fi paper and repository.\]](#)

[Missing modality learning] [\[SOURCE TO BE VERIFIED: Representative missing-modality learning survey or benchmark.\]](#)

[Reliability-aware fusion] [\[SOURCE TO BE VERIFIED: Representative reliability-aware or uncertainty-aware multimodal fusion paper.\]](#)

[WiFi-CSI HAR] [\[SOURCE TO BE VERIFIED: Representative WiFi-CSI human activity recognition paper.\]](#)

[mmWave sensing] [\[SOURCE TO BE VERIFIED: Representative mmWave human sensing paper.\]](#)

Table E.19: Coverage of local result files in the manuscript.

Source group	Representative files	Manuscript use	Reason
HPE student results	Student-VK and Student-NV combination CSVs; cross-scene/cross-subject CSVs	Main and appendix	Formal HPE evidence for all-combination robustness
HAR student results	Student-VK and Student-NV combination CSVs; cross-scene/cross-subject CSVs	Main and appendix	Formal HAR evidence for all-combination robustness
Reliability weights	HPE/HAR Student-VK reliability-weight CSVs	Main figure and table	Direct support for dynamic weighting behavior
Ablations	No-distillation, uniform-fusion, encoder and KD-variant outputs	Main/appendix selectively	Supports module analysis; extreme collapses are not promoted
Reproduced Fi/VK Super-Teacher	X- origin-XFI/Ori-HPE; origin-XFI/Ori-HAR outputs Super-Teacher/outputs	Baseline only Exploratory appendix	Prior-work reference, not our contribution Useful diagnostic but not strong enough as a main claim
XRF55 outputs	No usable result CSV/JSON/TXT/MD found during inspection	Excluded	No auditable result file available

[LiDAR human perception] [\[SOURCE TO BE VERIFIED: Representative LiDAR human perception paper.\]](#)

[Depth-based human perception] [\[SOURCE TO BE VERIFIED: Representative depth-based pose or action recognition paper.\]](#)

[Pose distillation] [\[SOURCE TO BE VERIFIED: Knowledge distillation for pose estimation.\]](#)

[Activity distillation] [\[SOURCE TO BE VERIFIED: Knowledge distillation for action recognition.\]](#)

[Multimodal transformer] [\[SOURCE TO BE VERIFIED: Representative multimodal transformer or token-fusion paper.\]](#)